ICT103 Programming

Topic 1
Introduction to Programming

CIHE Australia

Resources

Prescribed Texts

- McMullen, K., Matthews, E., & Parsons, J. J. (2022). Programming with Python (1st ed.)., Cengage Learning

Recommended Readings

- Lutz, M. (2013). Learning Python (5th ed.). O'Reilly Media, Inc

* Slides taken from Cengage resources

CENGAGE

Week 1: Lesson Learning Outcomes

- Understand the need for programming
- Explain job prospects and the role of a programmer in the industry
- Develop problem-solving skills and understand the problem-solving approach
- Understand algorithms, flowcharts and their significance in programming
- Learn why Python is widely used and its advantages over other languages
- Understand fundamental programming terminologies

What is Computer Science?

- A *computer* is a programmable machine that receives input, stores and manipulates data, and provides output in a useful format. *(Wikipedia)*
- *Computer science* is the study of the theoretical foundations of information and computation, and of practical techniques for their implementation and application in computer systems. *(Wikipedia)*
- Fields include AI, Cybersecurity, Networking, Data Science, and Software Engineering.
- Why do we need computers:
  - Automation: Replacing repetitive tasks (e.g. online banking)
  - Faster processing: Solving complex problems (e.g. weather forecasting)
  - Data Storage: Managing large amounts of information (e.g. Cloud Computing)
  - Communicate globally (e.g. Social media)

What is Programming?

- Programming = Writing instructions for a computer.

- "Programming is the act of writing or debugging a computer program." *(Wikipedia)*
- A *program* tell a computer **what to do and how to do it**.
- Without programming, computers are just expensive boxes!

- Example:
  - An ATM program checks your PIN before allowing withdrawals

Why do we need Programming?

- Computer programming is the lifeblood of modern life.
- Computers need precise instructions to work.
- *Programming helps us communicate with computers to solve real-world problems.*
- Everything from mobile apps to self-driving cars relies on programming.

Applications of Programming?

Web Development	Websites like Google, Facebook, Amazon
Artificial Intelligence and Machine Learning	Virtual assistants (Siri, Alexa)
Cybersecurity	Encryption, ethical hacking (banking security)
IoT and Smart Devices	Smart home automation (Alexa, Google Home)
Finance and Business Analytics	Stock market predictions, fraud detection
Gaming	Game engines (Unity, Unreal Engine)
Cloud Computing	Google Drive, AWS, Microsoft Azure

Role of a Programmer in the Industry

- *A programmer is responsible for designing, writing, testing and maintaining code.*

- **Develops** software and applications
- **Solves** real-world problems using logical thinking
- **Works** with teams (collaborates with designers, engineers, analysts)
- **Optimizes** processes to make business more efficient

- Skills needed:
  - Technical skills: Programming languages, debugging, problem-solving.
  - Soft skills: Logical thinking, teamwork, adaptability, patience.

Career Opportunities

- Programming skills open doors to various career opportunities.
- High demand. Every industry requires programmers.
- Global opportunities. Work remotely or in top tech companies.
- Well-paid roles. Salaries increase with experience.

Job Roles
Software Developer
AI Engineer
Cybersecurity Analyst
Data Scientist
Cloud Engineer

What does Programmer do?

- *A programmer's job is to sole problems through code*.
- Breaks problems into small, manageable parts.
- Uses logic and algorithms to find solutions.
- Translates real-world issues into working software.

What is Problem Solving in Programming?

- *Problem Solving* is the sequential process of analyzing information related to a given situation and generating appropriate response options.

- Problem Solving Process:
  - Understanding the problem
  - Analyzing the problem
  - Developing a plan
  - Implementing the plan
  - Testing and Debugging
  - Evaluating the solution

Steps in Problem Solving

- Define the problem
- Break it into smaller tasks
- Design an algorithm
- Code the solution
- Test and debug

- **Example**
- Scenario: Making tea

- Boil water
- Add tea leaves
- Pour into a cup
- Add sugar/milk
- Serve

* *This is like writing an algorithm!*

What is an Algorithm?

- Algorithm:
  - A series of steps for solving a problem or carrying out a task.
- Programming algorithm:
  - A set of steps that specify the underlying logic and structure for the statements in a computer program.
- Helps to break down complex problems.
- Makes problem-solving systematic and efficient
- Forms the foundation of programming logic

Representations

- Two ways:
  - Flow charts
  - Pseudocode

- Flow charts
  - Symbols convey different types of actions
  - Graphical representation of an algorithm
  - Uses symbols to show process flow

Flowchart

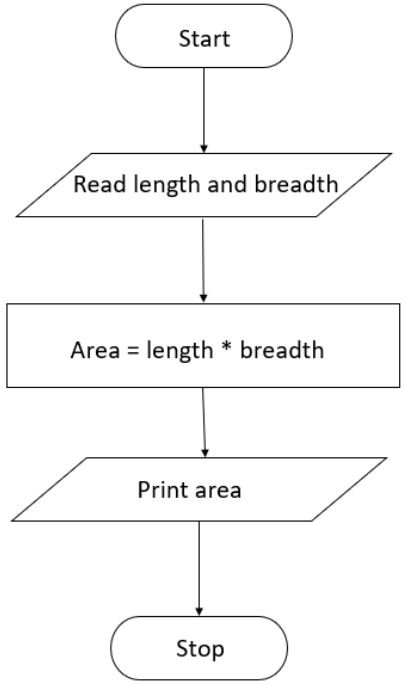
Symbol	Name	Function
	Start/End	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/ Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Pseudocode

- Pseudocode is text based – Plain English to express the logic and flow of the program
- A way of describing an algorithm without using any specific programming language
- No graphical or tabular syntax

Example

- Draw a flowchart and write a pseudocode to calculate the area of a rectangle
- Formula: Area = Length * Breadth

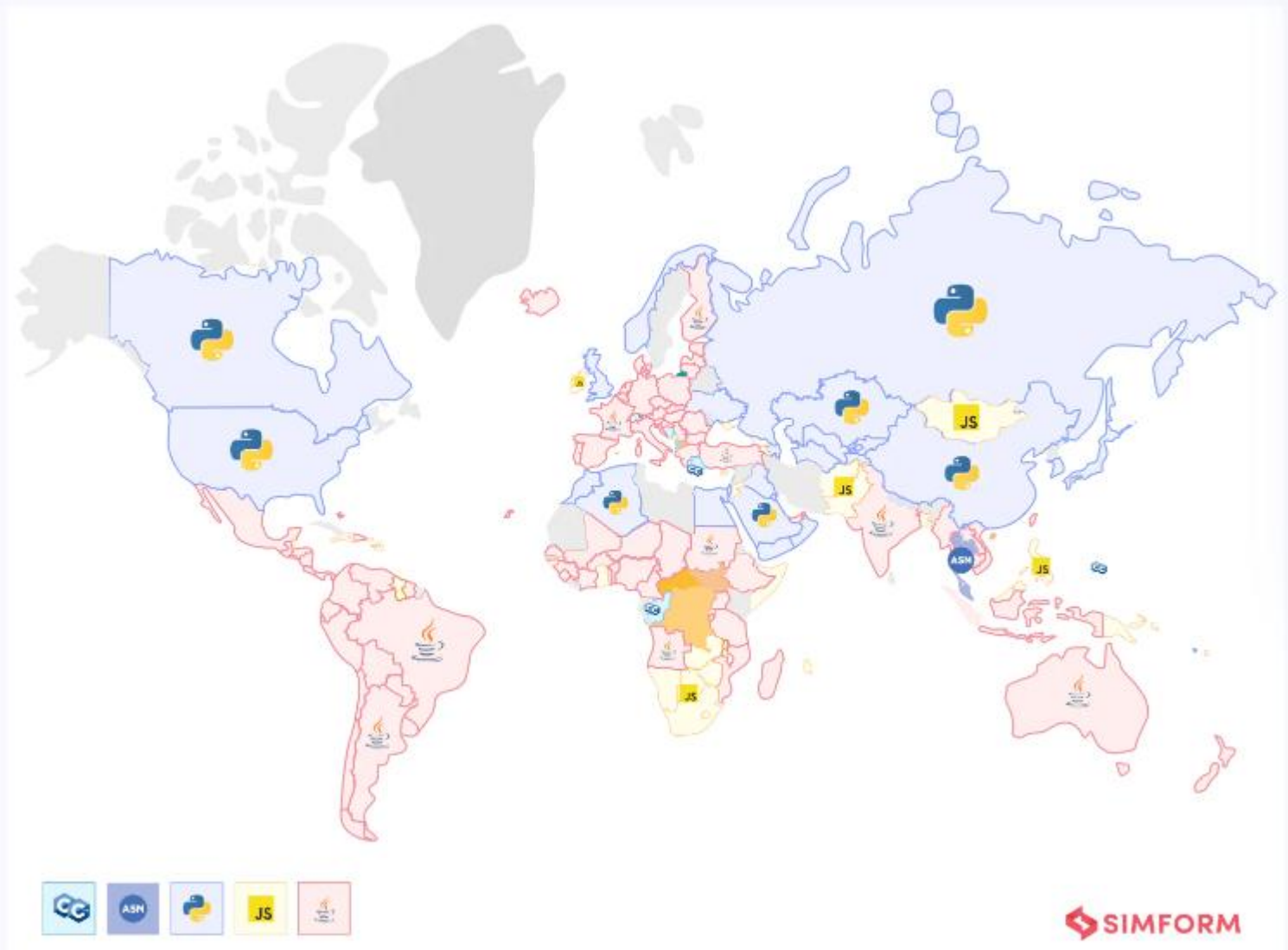
Pseudocode	Flow chart
Step 1: Start Step 2: Input length and breadth Step 3: area = length * breadth Step 4: print area Step 5: stop	

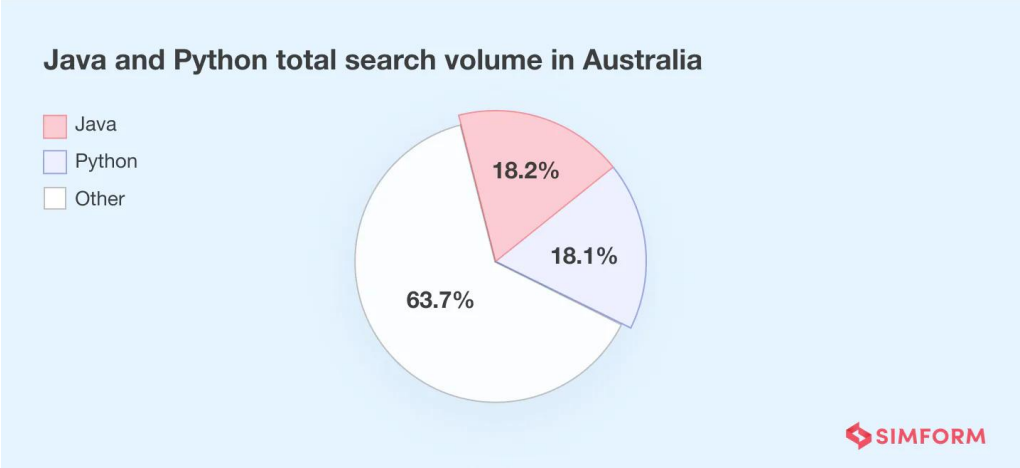
**Use Draw.io to draw flowcharts

Different Types of Programming Languages

- Low-level languages: Assembly, Machine code
- High-level languages: Python, Java C++
- Scripting languages: JavaScript, PHP

Programming Languages





Popular Programming Languages and Their Uses

Language	Usage
Python	AI, Data Science, Web Dev
Java	Mobile, Apps, Enterprise Software
C++	Game Development, System Software
JavaScript	Web Development
SQL	Database Management
C#	Game Development (Unity)

Why Python?

- Simple and Readable: Uses English-like syntax
- Versatile: Used in AI, web dev, automation and data science
- Community Support: Large community with vast libraries
- Cross-Platform: Works on Windows, MAC, and Linux

Feature	Python	Java	C++	JavaScript
Ease of Use	✅ Simple & easy	❌ More complex	❌ Harder syntax	✅ Easy for web dev
Speed	⚠️ Slower	✅ Fast	✅ Very fast	⚠️ Moderate
Applications	✅ AI, Web, Data Science	✅ Enterprise Apps	✅ Game Dev	✅ Web Apps

Simple Python Code

- Starting point for programmers

Hello World! in C++

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello World!" << endl;
    return 0;
}
```

Hello World! in Java

```
class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

Hello World! in Python

```
print("Hello World!")
```

Key Programming Terminology

- Syntax
  - A specific set of rules for the language
- Program Statements
  - Similar to English sentences
  - Commands to carry out program tasks
- Compiler
  - Preprocesses and converts all the statements into a binary file
- Interpreter
  - Converts source code into object code one statement at a time
- Syntax Errors
  - Misuse of language rules
  - A misspelled programming language word
- Logic Errors
  - Also called semantic errors
  - The program may run but provide inaccurate output
- Debugging
  - Finding and fixing errors in code

Next Week – What's Coming Up?

- Declare, initialize and manipulate variables
- Explain significance of primitive data types
  - Demonstrate operational skills with mathematical operators and compound operators
- Identify and manipulate character data and string data
  - Use built-in functions to manipulate strings
- Explain meanings of coercion and typecasting in the context of data types

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